Ahsanul Hoque

Founder and CEO, Livora | Software Engineer (Full-Stack, AI and Embedded Systems)

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Summary

Computer Science undergraduate (University of Chittagong, expected 2026) and Founder/CEO of Livora, a full-stack AI-powered healthcare platform that won 2nd place and an incubation package at the ITBI Student Startup Pitch Fest 2026 (CUET). Ships production-grade full-stack and applied-AI systems across healthcare, e-commerce, embedded/IoT, and data engineering. Comfortable across the stack: React/TypeScript frontends, Node.js/Express APIs, SQL data modeling, LLM/RAG integration, and embedded systems (ESP32, LoRa).

Experience

Founder and CEO – Livora (Self-employed) **May 2026 – Present**

Chattogram, Bangladesh (Hybrid)

- Lead product, engineering, and business strategy for Livora, an AI-powered healthcare platform serving patients, doctors, lab technicians, and hospital administrators
- Won 2nd place at the ITBI Student Startup Pitch Fest 2026 (CUET), securing 6 months of free incubation space, mentoring, and startup-ecosystem support worth BDT 100,000
- Conducted an AI training workshop on the platform for practicing surgeons in Chittagong

Skills

Languages: Python, C++, TypeScript, JavaScript, SQL

Web and Backend: React, Node.js, Express, Prisma, Sequelize, REST APIs, Socket.IO

AI / LLM: LLM Integration (Google Gemini), RAG (Retrieval-Augmented Generation), Prompt Injection Defense, Langfuse (LLM Observability), OCR (Tesseract), Speech-to-Text (Deepgram)

Data and Infrastructure: PostgreSQL, MySQL, Redis, Docker, HL7 / FHIR / DICOM (healthcare interoperability), Cloudinary, Pino, Sentry

Embedded and IoT: ESP32, Arduino, FreeRTOS, LoRaWAN, Raspberry Pi, ThingSpeak

Other: Data Structures and Algorithms, Competitive Programming, Blockchain Fundamentals, Project Management

Projects

Livora – AI-Powered Healthcare Platform *Oct 2025 – Present*

React, TypeScript, Node.js, PostgreSQL, LLM/RAG, Socket.IO

- Building a full-stack clinical platform covering appointment booking, voice-driven prescriptions, lab order/result management with payments, DICOM/FHIR and HL7 ingestion, and multi-disciplinary team (MDT) video consultations
- Implemented a persistent AI memory system and a RAG-based health assistant with medicine safety checks, risk stratification, and prompt-injection defenses; instrumented with Langfuse and Sentry

ResCris – Long-Range Emergency and Crisis Management Network *2025*

ESP32, LoRa (SX1278), GPS, Embedded Systems

- Built an internet-independent emergency communication system using LoRaWAN for disaster response, with real-time SOS reporting, GPS-based tracking, and a multi-gateway architecture
- Presented at Chittagong Science Carnival 4.0 with a teammate

Aurora Gadgets – Full-Stack E-Commerce Platform *Apr 2026 – May 2026*

React, TypeScript, Express, Prisma, MySQL, Docker

- Built a complete storefront and admin system covering products, categories, brands, cart, and checkout with category-based product attributes
- Containerized the application with Docker for reproducible deployment

Agri Supply Chain Transparency Platform *2025*

Blockchain

- Designed a blockchain-based platform to make farm-to-consumer pricing transparent and verifiable, aimed at closing the margin gap between farmer income and consumer prices
- Runner-up, Hackathon Segment, Chittagong Science Carnival (hosted by CUSS)

Warehouse Management and Billing System *Freelance*

Node.js, MySQL

- Built an inventory, billing, and sales-analytics system for a retail client

Sales Data Warehouse and BI Reporting *Aug 2025 – Sep 2025*

SQL, Python, Jupyter Notebook

- Designed and built a modern SQL data warehouse from scratch, consolidating sales data via ETL and dimensional modeling
- Built interactive BI dashboards with Python (Pandas, Matplotlib, Seaborn) for trend and segmentation analysis

AgroEdge AI – Precision Agriculture Platform *Jan 2026 – Apr 2026*

Python, Raspberry Pi, ThingSpeak

- Built an AI-first precision-agriculture stack: data pipeline, irrigation model training, edge inference, and a Raspberry Pi dashboard integrated with ThingSpeak

WaifOS – Multitasking OS for ESP32 *Mar 2025 – May 2025*

C++, Arduino, FreeRTOS

- Built a lightweight multitasking operating system for ESP32 featuring an app menu, display rendering, and input handling
- Implemented WiFi connectivity and concurrent task scheduling on constrained embedded hardware

Education

University of Chittagong – B.Sc. in Computer Science and Engineering *2022 – 2026 (Expected)*

Govt. City College, Chattogram – Higher Secondary Certificate, Science (GPA 5.00) *2019 – 2022*

Chittagong Collegiate School – Secondary School Certificate, Science (GPA 5.00) *2013 – 2019*

Awards and Achievements

- 2nd Place, ITBI Student Startup Pitch Fest 2026, CUET – for Livora (incl. incubation support)
- 1st Place, Robo Soccer, EEE Fest 2025, University of Chittagong
- Runner-up, Hackathon Segment, Chittagong Science Carnival (hosted by CUSS)
- 1st Runner-up, Robo Soccer, Bangabandhu Innovation Fair 2024, Premier University
- Champion, ETE DecaHz – SDG-focused idea competition (algae-based circular economy model)
- 2nd Runner-up, Intra-Department Programming Contest, 22nd Batch

Leadership and Activities

- President, N-Cube (Science Club), Chittagong Collegiate School
- Joint Secretary, Debating Society, Chittagong Collegiate School